llmXive Automated Review of OmniOpt: Taxonomy, Geometry, and Benchmarking of Modern Optimizers

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive taxonomy and benchmark of modern optimizers. However, several claims rely on citations to papers with future dates (2026) or methods that appear to be hypothetical or not yet widely recognized. Specifically, the benchmark results for RMNP, SPlus, SAC, and SGG are based on citations to 2026 preprints or methods that may not be fully established. This raises concerns about the reproducibility and validity of the benchmark claims. The authors should verify the existence and publication status of these methods and ensure that the benchmark results are based on real, reproducible experiments. Additionally, the citation for '8-bit Adam' in Section 2.1 should be clarified to ensure it matches the cited work's scope.

Action items.

- [**writing**] The paper presents a comprehensive taxonomy and benchmark of modern optimizers. However, several claims rely on citations to papers with future dates (2026) or methods that appear to be hypothetical or not yet widely recognized. Specifically, the benchmark results for RMNP, SPlus, SAC, and SGG are based on citations to 2026 preprints or methods that may not be fully established. This raises concerns about the reproducibility and validity of the benchmark claims. The authors should verify the exi

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure effectively visualizes the taxonomy and timeline of optimizer evolution, but the x-axis typography is cluttered, with specific year labels merging into time ranges, which reduces readability.

Figure 2

Figure 2 effectively illustrates the universal meta-pipeline for an optimizer step with clear stage progression and internal operations. Minor improvements could be made to clarify the temporal iteration flow and the explicit connection between transformer components and trainable parameter groups.

Figure 3

Figure 3 is a clear and well-organized taxonomy diagram that effectively visualizes the relationships between various T1 optimizer variants. The use of color-coded sections, directional arrows, and detailed text boxes for specific algorithms (like AdamW and AdEMAMix) makes the mechanism overview easy to follow and consistent with the caption's description.

Figure 4

Figure 4 is a clear and well-organized taxonomy schematic that effectively visualizes the three T2 matrix-level routes (Spectral Orthogonalization, Kronecker-Factored Preconditioning, Low-Rank Subspace Projection) and their constituent methods. The color-coding, hierarchical layout, and internal annotations (S1-S5) are legible and align perfectly with the provided caption.

Figure 5

Figure 5 is a clear and well-organized taxonomy schematic that effectively categorizes T4 optimizers by their memory reduction mechanisms. The visual hierarchy, color-coding, and internal text descriptions (S1-S5) align perfectly with the caption's description of the four sub-groups.

Figure 6

Figure 6 is a clear and well-organized taxonomy schematic that effectively categorizes T5 curvature-aware and geometric regularization methods into four distinct sub-groups. The visual hierarchy, color-coding, and internal text descriptions (S1-S5) are legible and align perfectly with the caption's description of the intervention points.

Figure 7

The figure visually presents Pareto frontiers with color-coded families and star-marked frontier members, but incorrectly labels non-frontier methods (e.g., Lion) as frontier members, contradicting the definition of Pareto optimality. Axis labels are clear but slightly redundant.

Figure 8

Figure 8 presents a clear heatmap of optimizer metrics, but the colorbar legend is ambiguously labeled. The 'Time/Mem baseline' label is misleading because the color scheme applies to PPL as well, and the legend does not explicitly state that green indicates better performance (lower values) for all three metrics relative to AdamW. This could lead to misinterpretation of the data.

Figure 9

Figure 9 presents rank stability data clearly, but the caption incorrectly describes the use of 'hatching' for architecture in panel (c) when the plot actually uses color intensity/opacity. Additionally, the inverted y-axis in panels (a) and (b) lacks explicit annotation to indicate that lower rank numbers (better performance) are at the top.

Figure 10

The figure presents a heatmap of gradient-norm stability but suffers from a mismatch between the raw numerical values in the cells and the normalized colorbar scale, alongside ambiguous x-axis labels that fail to define the specific architectures tested.

Figure 11

The figure effectively visualizes learning-rate robustness across optimizers with clear color coding and data points. However, there is a minor notation inconsistency between the caption ($s_L R$) and the panel labels (s), and the legend symbol description could be better aligned with the caption's terminology.

Figure 12

The figure presents a family-level summary heatmap but fails to map the displayed columns to the O1–O6 objectives mentioned in the caption. Additionally, the lack of defined units or directionality (higher/lower is better) for the specific metrics makes the color-coded 'best/worst' interpretation ambiguous.

Action items.

- [**writing**] Figure 1: The x-axis timeline is cluttered and inconsistent; the '2014' label is merged with '2015 - 2019', and the '2020' label is merged with '2020 - 2023', making the specific year boundaries difficult to parse.
- [**writing**] Figure 1: The x-axis labels '2014' and '2020' are visually merged with adjacent time ranges (e.g., '2014 2015 - 2019'), creating ambiguity about the start of the 'Stage II' and 'Stage III' periods.
- [**writing**] Figure 2: The top-left 'Iterations' arrow points to the S0 block but lacks a clear label or connection to the 'Training Iteration t' box, creating ambiguity about the temporal flow.
- [**writing**] Figure 2: The 'Transformer Blocks' section lists 'Self-Attn' and 'MLP / FFN weights' but does not explicitly link these to the 'trainable parameter groups' arrow feeding into S0, which could confuse readers about the source of gradients.
- [**science**] Figure 7: The caption states 'stars mark frontier members,' but the left plot shows 'Lion' (green star) and 'GaLore' (red star) as stars, while 'APOLLO' (purple star) is also a star. However, 'Lion' is clearly dominated by 'GaLore' and 'APOLLO' in both PPL and time/memory, so it should not be on the Pareto frontier. This misrepresents the frontier definition.
- [**writing**] Figure 7: The x-axis label 'Per-step time (ms)' on the left plot and 'Optimizer-state memory (GB)' on the right plot are clear, but the y-axis label 'C4 val PPL (1B)' is repeated

identically on both plots without clarifying that the left is time and right is memory — though this is minor since the x-axes differ.

- [**science**] Figure 8: The colorbar legend is inverted relative to the data and caption. The caption states 'Green is favorable' (lower PPL/Time/Memory), but the colorbar labels 'better than AdamW' at the top (green) and 'worse than AdamW' at the bottom (red). However, the data shows AdamW (PPL 14.48) is colored green, while methods with better PPL (e.g., Adan 14.35) are also green, and methods with worse PPL (e.g., AdaBelief 16.79) are red. The issue is that the colorbar labels 'Time/Mem baseline' and 'same
- [**writing**] Figure 8: The colorbar legend has ambiguous labels. The top label 'Time/Mem baseline' is unclear because the heatmap includes three metrics (PPL, Time, Memory), but the label suggests the baseline applies only to Time and Memory. The middle label 'same as AdamW' is placed on the colorbar, but it is not clear if this refers to the color value or the metric value. The bottom label 'worse than AdamW' is clear, but the overall legend design is confusing. The colorbar should be relabeled to explicitl
- [**writing**] Figure 9 caption: states 'hatch = architecture' for panel (c), but the rendered plot uses color intensity (opacity) to represent architecture, not hatching patterns.
- [**science**] Figure 9: The y-axis in panels (a) and (b) is inverted (1 at top, 12 at bottom) to indicate '1 = best', but the axis labels are not explicitly marked as inverted or 'best at top', which can be confusing for standard plot reading.
- [**writing**] Figure 9: The x-axis labels in panels (a) and (b) are stacked vertically (e.g., 'Tr++' over '340M'), which is readable but could be improved by rotating or spacing them to avoid crowding.
- [**writing**] Figure 9: The legend in panel (b) is placed inside the plot area and partially obscures the data lines; moving it outside or making it semi-transparent would improve readability.
- [**writing**] Figure 9: The x-axis labels in panel (c) are rotated at an angle, which is good for readability, but some labels (e.g., 'MARS-Shampoo') are still slightly cut off or hard to read at the edges.
- [**writing**] Figure 9: The caption mentions 'hatch = architecture' for panel (c), but the plot uses color intensity (opacity) instead of hatching, creating a mismatch between description and visual representation.
- [**science**] Figure 10: The heatmap cells display raw GNormCV values (e.g., 111, 161) while the colorbar scale is normalized (0.3–111.6), creating a misleading visual mapping where the highest numerical value (161) is colored identically to the scale's maximum (111.6).
- [**writing**] Figure 10: The x-axis labels ('340m', '1b') are ambiguous and do not specify the architecture or scale they represent, which is critical for interpreting the 'across architectures' claim in the caption.

- [**writing**] Figure 11: The caption defines the sensitivity score as '$s_L R$', but the panels display 's'; align the notation in the caption with the figure labels.
- [**writing**] Figure 11: The legend uses the symbol ' ' for 'Best LR', but the caption states 'The star marks the tuned learning rate'; clarify if 'Best LR' and 'tuned learning rate' are synonymous.
- [**science**] Figure 12: The caption states the profile covers objectives O1–O6, but the heatmap only displays five columns (Performance, Efficiency, Stability, Robustness, Generalization). The mapping between these labels and the O1–O6 objectives is not provided in the caption or the figure, making it impossible to verify the data against the claimed metrics.
- [**science**] Figure 12: The colorbar indicates that green is 'best' and red is 'worst', but the underlying data values are not normalized or defined. For example, 'Performance' shows 14.35 (green) while 'Robustness' shows 23.0 (red); without knowing if lower or higher is better for each specific metric, the color mapping appears arbitrary and potentially misleading.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a specialized audience, but it relies on a few internal shorthand conventions and symbols that are introduced in one section and then used extensively in others without immediate re-expansion or definition.

Specifically, the symbol `S_{opt}` in the memory equation (Section 2.1) appears without a definition, creating a minor barrier for a reader not intimately familiar with the authors' specific notation for optimizer state counts. Similarly, the acronym "LMO" is used in Section 3.1 before being explicitly expanded in that specific section, despite being defined earlier in the Introduction; for a self-contained reading experience, it should be expanded at its first occurrence in the body of the paper where the mathematical framework is detailed.

The notation `u^\sharp` in Equation 2 is also used without a definition of the dual map operation, which is a standard but specific piece of convex analysis notation that might not be immediately obvious to a reader from a purely engineering or applied ML background. Finally, the heavy reliance on shorthand labels like "S0", "S1", "T1", etc., throughout the later sections (Sections 4 and 5) assumes the reader has the taxonomy table or Section 3.1 open. While not fatal, adding brief parenthetical expansions at the first use of these shorthands in each new major section would significantly improve flow and accessibility for an adjacent-field PhD.

Action items.

- [**writing**] The paper is generally well-structured for a specialized audience, but it relies on a few internal shorthand conventions and symbols that are introduced in one section and then used extensively in others without immediate re-expansion or definition. Specifically, the symbol S_{opt} in the memory equation (Section 2.1) appears without a definition, creating a minor barrier for a reader not intimately familiar with the authors' specific notation for optimizer state counts. Similarly, the acronym "

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is generally sound, establishing a taxonomy and then validating it with benchmarks. However, there are specific breaks in the logical chain between the textual claims in Section 5.2 (Mechanistic Ablation of Muon) and the data presented in Table 5 (tab_muon_cross_validation).

First, the text asserts that "removing AdamW's second moment collapses PPL to 70.74." This is a specific quantitative claim used to justify the importance of the second moment. However, Table 5, which is cited as the source for this ablation study, does not list a "No Second Moment" condition, nor does it contain the value 70.74. The table only compares Standard Muon, LR Scaling, Nesterov, and their combination. The conclusion (the collapse to 70.74) does not follow from the premises (the data in Table 5) because the necessary data point is missing from the cited evidence. This is a non-sequitur; the reader cannot verify the claim from the provided table.

Second, the text states that adding Newton-Schulz (NS) orthogonalization "recovers to 16.86." In the same section, the text identifies "Standard Muon" as the baseline containing NS. Table 5 reports the PPL for "Standard Muon" at 350M as 16.60. The discrepancy between the text's value (16.86) and the table's value (16.60) for the same configuration creates an internal inconsistency. While this could be a typo, it breaks the logical consistency of the argument if the text is describing a specific run that differs from the table's reported best run without clarification.

Third, the text concludes that "LR scaling and Nesterov provide secondary gains" and that "gains stack." While this holds for the Transformer architecture in Table 5, the table explicitly shows that for the Gated DeltaNet (GDN-340M) architecture, the "Both combined" configuration (24.12) performs worse than "Symmetric LR Scaling" alone (24.02). The text's general statement that gains stack or that these are secondary gains contradicts the specific evidence in the table where the combination is detrimental. The argument overgeneralizes from the Transformer results to a broader claim that is falsified by the GDN data presented in the same table.

These issues require the authors to either align the text with the table values (correcting the numbers or the description of the ablation) or to explicitly explain the discrepancy (e.g., "in a separate run not shown in Table 5..."). The current state leaves the reader unable to trace the specific numerical claims back to the provided evidence.

Action items.

- **[science]** Section 5.2 claims 'removing AdamW's second moment collapses PPL to 70.74' in the Muon ablation, but Table 5 (tab_muon_cross_validation) does not contain a 'No Second Moment' row or the value 70.74. The text describes a specific ablation condition that is not represented in the cited table, creating a non-sequitur between the claim and the evidence.
- **[writing]** Section 5.2 states 'adding Newton–Schulz (NS) orthogonalization recovers to 16.86,' yet Table 5 lists the 'Standard Muon' (which includes NS) PPL as 16.60 at 350M. The text's specific value (16.86) contradicts the table's value (16.60) for the same configuration without explanation.
- **[writing]** Section 5.2 claims 'LR scaling and Nesterov provide secondary gains,' but Table 5 shows that for Gated DeltaNet (GDN-340M), the 'Both combined' configuration (24.12) is worse than 'Symmetric LR Scaling' alone (24.02). The text's generalization of 'gains' contradicts the specific data point in the table where the combination degrades performance.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper presents a rigorous benchmark, but the framing in the Abstract and Conclusion occasionally overstates the universality of the findings. While the body text (Section 5) carefully qualifies results by model size, context length, and architecture, the concluding remarks present specific observations as general laws.

Specifically, the claim that the benchmark "validates" a universal lack of dominance and that selection "must" match geometry implies a scope (all possible optimizers and domains) that exceeds the evidence (24 optimizers, 4 architectures, 2 domains). Similarly, labeling the failure of state compression in long contexts and the sensitivity of spectral methods as "empirical regularities" elevates specific trends from the tested subset to universal principles.

The paper's own Limitations section (Section 6.1) correctly identifies these boundaries, but the Conclusion re-opens the scope without the necessary qualifiers. To align the rhetoric with the evidence, the authors should:

1. Replace strong modal verbs ("validates", "must") with hedged language ("suggests", "should") in the Abstract and Conclusion.
2. Explicitly scope claims about "regularities" to the specific method families (T2, T4) and architectures tested.
3. Ensure the conclusion reflects the protocol-relative nature of the findings rather than asserting them as absolute truths.

These are fixable writing adjustments that will ensure the paper's confidence level matches the actual scope of its experimental evidence.

Action items.

- [**writing**] Abstract/Conclusion: Replace 'validates that no single optimizer dominates' with 'suggests no single optimizer dominates under tested protocols.' Qualify 'selection must match' to 'selection should match within these constraints' to avoid universal claims beyond the 24 optimizers and 2 domains tested.
- [**writing**] Conclusion: Rephrase 'Two empirical regularities' to 'Two observed trends in our study.' Specify that 'aggressive state compression is rank-bounded' applies to the tested T4 methods (APOLLO, AdaFactor) and 'spectral geometry is architecture-conditional' refers to the specific T2 instances (Muon, SOAP) evaluated.
- [**writing**] Section 5.2.5: Change 'Optimizer performance is strongly architecture-dependent' to 'Performance varied across the tested vision backbones.' Scope the claim to the three specific architectures (ResNet50, DeiT-S, CAFormer-S12) to prevent overgeneralization to untested model families.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a taxonomy and benchmarking study of modern optimization algorithms for deep learning. From a safety and ethics perspective, the work is low-risk. The research focuses on algorithmic efficiency, convergence properties, and memory usage of optimizers (e.g., AdamW, Muon, SOAP, Lion) applied to standard benchmarks (C4, FineWeb-Edu, CIFAR100).

There are no indications of dual-use capabilities that lower the barrier to harmful activities (e.g., automated vulnerability discovery, biological synthesis, or persuasive disinformation generation). The methods described are standard mathematical operations for training neural networks and do not constitute operational exploits or attack vectors.

The data sources cited (C4, FineWeb-Edu, CIFAR100) are public, widely used datasets in the ML community. The paper does not appear to use scraped data in violation of terms of service, nor does it release any datasets containing personally identifiable information (PII) or sensitive human subject data. No human subjects were involved in the experiments (no surveys, interviews, or behavioral logs), so IRB/consent statements are not required.

The paper does not disclose any conflicts of interest, but given the nature of the work (benchmarking existing open-source algorithms), there is no immediate evidence of undisclosed commercial bias that would constitute a safety or ethical violation. The authors acknowledge limitations regarding protocol sensitivity and scope, which is appropriate.

No specific, foreseeable risks of harm were identified that require mitigation or disclosure beyond what is standard for this type of algorithmic benchmarking research.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents an extensive benchmark of 24 optimizers across multiple scales and architectures, but the evidentiary strength of the central claims is compromised by a lack of variance reporting and potential confounds in hyperparameter tuning.

First, the headline results in Section 4.2 (Tables 1, 2, and 3) are presented as definitive rankings based on single training runs. For instance, the claim that APOLLO achieves a PPL of 13.53 at 1B parameters, significantly outperforming AdamW (14.48), is reported without any measure of stability (standard deviation, standard error, or confidence intervals). In large-scale LLM training, performance can vary by 0.5–1.0 PPL points across different random seeds due to initialization and stochasticity. Without reporting results across at least 3–5 seeds, it is impossible to distinguish a genuine algorithmic improvement from a lucky seed. The current design cannot rule out that the reported "best" optimizers are simply the result of favorable random initialization.

Second, the dramatic failure mode attributed to APOLLO in long-context training (Section 4.2, "Scenario Sensitivity") appears confounded by hyperparameter settings. The text claims APOLLO collapses (PPL 35.40) while AdamW remains stable, but the hyperparameter table (Appendix) reveals that APOLLO was trained with a learning rate of 0.01 at 1B scale, whereas AdamW used 0.001. This tenfold difference in learning rate is a massive confound; the "collapse" could easily be a result of the optimizer being pushed into an unstable regime by an aggressive learning rate rather than an intrinsic inability to handle long contexts. To support the claim that APOLLO is "rank-bounded" or "fails" in long contexts, the authors must demonstrate this failure under a matched or carefully tuned learning rate schedule comparable to the baseline.

Finally, the mechanistic ablation of Muon (Section 4.2) attributes performance gains primarily to the Newton-Schulz (NS) orthogonalization step. However, the ablation compares the full Muon method against a baseline that likely differs in both the orthogonalization operator and the momentum state evolution (S3). The Muon paper specifically modifies how momentum is accumulated in the matrix space. Without a control run that keeps the Muon-specific momentum state evolution but removes the NS orthogonalization (replacing it with identity), the design cannot isolate the contribution of the geometric operator from the state evolution changes. The reported gains might be driven by the momentum structure rather than the orthogonalization itself.

To strengthen the evidence, the authors should: (1) report mean and standard deviation over multiple seeds for all key benchmark results; (2) re-run the long-context APOLLO experiments with a learning rate schedule matched to AdamW to rule out tuning artifacts; and (3) refine the Muon ablation to isolate the NS operator from the momentum state evolution.

Action items.

- **[writing]** The paper presents an extensive benchmark of 24 optimizers across multiple scales and architectures, but the evidentiary strength of the central claims is compromised by a lack of variance reporting and potential confounds in hyperparameter tuning. First, the headline results in Section 4.2 (Tables 1, 2, and 3) are presented as definitive rankings based on single training runs. For instance, the claim that APOLLO achieves a PPL of 13.53 at 1B parameters, significantly outperforming AdamW (14.48)

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the benchmark results in this paper is insufficient to support the definitive ranking and "winner" claims made in the text. While the experimental design (Section 5.1) is extensive, the analysis of the resulting data lacks the necessary rigor to distinguish signal from noise.

The primary issue is the reporting of **point estimates without uncertainty**. Throughout Section 5.2 and the Appendix tables (e.g., Tables 3-10), results are presented as single numbers (e.g., "13.53 PPL", "80.53% accuracy"). In stochastic deep learning training, performance varies significantly across random seeds. Reporting a single run (or an unreported average) as a fixed property of an optimizer is statistically unsound. The paper claims APOLLO is the "best" at 13.53 PPL compared to AdamW's 14.48, but without standard deviations or confidence intervals, it is impossible to know if this 0.95 difference is real or a lucky seed. The field standard for such benchmarks is to report mean ± standard deviation over at least 3-5 seeds.

Furthermore, the paper makes numerous **comparative claims** (e.g., "SOAP holds top PPL in 7/8 scenarios," "Muon improves DeiT-S by >5 percentage points") without addressing the **multiple comparisons problem**. With 24 optimizers tested across 4 architectures and 2 scales, the authors are effectively running hundreds of hypothesis tests. Highlighting the "best" performers without correcting for multiplicity (e.g., via Bonferroni or Benjamini-Hochberg) guarantees that some "significant" findings are false positives. The text uses terms like "best" and "wins" as if they are statistically proven facts, but no p-values, effect sizes, or confidence intervals are provided to back them up.

Finally, the **ablation study** in Section 5.2.8 (Muon) reports specific PPL drops (e.g., "70.74" vs "16.86") but again provides no variance estimates. While the magnitude of the drop is large, the lack of replication data prevents a formal assessment of the stability of these gains.

To fix this, the authors must either re-run experiments with multiple seeds to compute uncertainty metrics or, if re-running is not feasible, explicitly qualify all results as "single-run observations" and remove definitive claims of superiority. The current presentation parades point estimates as exact truths, which misleads the reader about the reliability of the findings.

Action items.

- **[science]** Section 5.2 and Tables 3-10 report single-point performance metrics (e.g., '13.53 PPL', '80.53%') without any measure of uncertainty (SD, SE, or CI) or mention of the number of random seeds used. In deep learning benchmarking, single-run results are indistinguishable from noise. Report mean ± SD over at least 3 seeds for all headline numbers, or explicitly state that results are from a single run and treat them as illustrative rather than definitive.
- **[writing]** Section 5.2.1 and Table 1 claim 'best' or 'worst' performance (e.g., APOLLO at 13.53 vs AdamW at 14.48) based on point estimates. Without reported variance or a statistical test (e.g., paired t-test or bootstrap), these differences cannot be distinguished from random fluctuation. Add error bars to figures and report statistical significance or effect sizes for all comparative claims.
- **[science]** The paper compares 24 optimizers across multiple architectures and scales (Section 5.2), effectively performing dozens of pairwise comparisons. However, no correction for multiple comparisons (e.g., Bonferroni, Holm, or FDR) is mentioned when declaring 'winners' or 'tiers'. This inflates the false-positive rate. Apply a correction method to the reported p-values or rephrase claims to avoid implying statistical significance where none was tested.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a dense, information-rich survey and benchmark. The prose is generally technical and precise, but several structural and syntactic issues create friction for the reader, requiring re-reading to parse the intended meaning or locate key claims.

The most immediate issue is a punctuation error in Section 3.1 (Universal Meta-Pipeline), where item S4 contains a stray closing parenthesis ("S4 (Reconstruction):)"). While minor, such errors disrupt the visual rhythm of a structured list. More significantly, the introduction of the "Four-Axis Decomposition" in Section 3.2 relies on an equation (Eq. 2) that uses undefined notation (D_ρ, $u^\sharp$) without immediate context. The reader must hunt for definitions or guess the variables, breaking the flow. A brief inline definition or a preceding sentence clarifying the LMO notation would resolve this.

Structurally, some paragraphs bury their main points. In Section 5.1 (T1), the claim that "T1's central commonality is coordinate-wise invariance" appears mid-paragraph after the equation. This is the paragraph's thesis and should lead the section to orient the reader immediately. Similarly, in Section 5.2, the description of Muon's role is tacked onto the end of a list of methods; separating this into a distinct sentence would improve clarity.

Finally, the "Technique-Level Lessons" in Section 6 attempts to present four distinct categories (Benefit carriers, Limited returns, etc.) within a single block of text. Without clear list formatting or paragraph breaks, the reader must work to distinguish these distinct lessons. Ensuring these are visually separated in the final typeset version is crucial for readability.

Overall, the paper is scientifically dense but requires light editing to ensure the prose delivers its complex arguments as smoothly as the science intends.

Action items.

- [**writing**] The manuscript presents a dense, information-rich survey and benchmark. The prose is generally technical and precise, but several structural and syntactic issues create friction for the reader, requiring re-reading to parse the intended meaning or locate key claims. The most immediate issue is a punctuation error in Section 3.1 (Universal Meta-Pipeline), where item S4 contains a stray closing parenthesis ("S4 (Reconstruction):)"). While minor, such errors disrupt the visual rhythm of a structure